Hang Xu
35414022

1. Declaration

I,**Hang Xu**, declare that this assignment, titled **Urban greening, tree-planting & biodiversity**, is my own original work and has not been copied from any other source except where explicitly acknowledged. I have not engaged in plagiarism, collusion, or any other form of academic misconduct in the preparation and submission of this assignment. All sources of information and data used in this assignment have been properly cited and referenced in accordance with the prescribed guidelines. I have not used unauthorized assistance in the preparation of this assignment and have not allowed any other student to copy my work. I am aware that any breach of academic integrity may result in disciplinary action as per the policies of Monash University, which may include failing this assignment or the course, and further academic penalties.

Signature: Hang Xu Date: 16/10/2026

2. Github Check

Enter your Github details here.

Github Username *Enter your username here*	hxuu0127-eng
Repository Shared? *Have you started and shared your assignment repository with your tutor yet?*	https://github.com/hxuu0127-eng/FIT5032-Assignment3-GreenLink

3. Self-Evaluation

Rate your performance for each criteria. Put a ✓(tick) in the box where you think your work belongs.

Criteria	Exceeds Expectations	Meets Expectations	Needs Improvement	Fail to meet expectations
BR (D.1): External Authentication	√			
BR (D.2): Email		√		
BR (D.3): Interactive Table Data	√			
BR (D.4): Deployment to the	√			

Cloud				
BR (E.1): Cloud Functions	√			
BR (E.2): Geo Location	√			
BR (E.3): Accessibility	√			
BR (E.4): Export	√			
BR (F.1): Innovation	√			

4. Screen Recording of BRs

Create a 6-10 minute video showing your advanced web application in action! Upload this video to your Google Drive and put the link here (ensuring that you have updated the access list so its not private).

https://drive.google.com/file/d/1RxKLsdlLtR1RI0-eF7Wp5yVSynnKpWfq/view?usp=sharing

5. BR F.1 - Innovative Features

If you have implemented BR F.1, list your choices below and a brief description of how you implemented it.

1	Smart Multi-Activity Booking	Implemented multi-activity booking using Firebase Cloud Functions, with automatic schedule conflict detection, capacity checks, and booking cancellation support.	Add a waiting list for fully booked activities and automatic email reminders before upcoming events.
2	Smart Activity Recommendations	Developed a Vue-based recommendation component to suggest relevant environmental activities based on user preferences and activity information.	Introduce machine learning to provide more personalised recommendations based on booking history, participation patterns, and user feedback.
3	Personal Environmental Impact Tracker	Implemented an environmental impact dashboard tracking	Extend the dashboard with estimated carbon savings, downloadable

		completed activities and volunteer hours through administrator-verified booking records stored in Firestore.	participation certificates, and monthly environmental impact reports.
4	Green Challenges & Achievements	Created five environmental challenges with progress tracking and automatic badge unlocking based on verified participation and volunteer hours.	Introduce seasonal challenges, community leaderboards, and additional achievement levels to encourage long-term participation in environmental activities.

6. Reflections: Challenges

What has been the most challenging part of this assignment for you? How has this stretched you as a programmer?

I think the most challenging part of this assignment for me are Firebase Cloud Functions and Firestore. When I was developing the multi-activity booking system, I had to handle booking conflicts, activity capacity, and real-time data updates. And I also faced issues with Firebase permissions. To overcome these challenges, I tested different scenarios, debugged errors, and improved both frontend and backend logic.

7. Declaration: Additional Help

Name	Description
Learning Vue.js and Firebase	I used ChatGPT to make it give me some Vue.js concepts, Firebase Authentication, Firestore, and Cloud Functions during the development of my GreenLink Melbourne application.
Testing and Debugging	I used ChatGPT to discuss testing scenarios for booking validation, geolocation, interactive tables, and data export functionality.
Brainstorming Innovative Features	I used ChatGPT to get some ideas like activity recommendations, environmental impact tracking, and gamification features. All of them is improving user engagement.

8. Learning Summary

8.1 Coverage of the Learning Outcomes

By this assignment, I have learned to use Vue.js to develop responsive and interactive web application. And I also gained practical experience like Firebase Authentication, Firestore, and Cloud Functions. What's more, I

implemented interactive tables, geolocation features, data export, and cloud deployment, so all of them could make me understand the frontend and backend technologies.

8.2 Reflect on your learning and discuss these areas (list and explain)

- **The most important thing I have learnt :** Integrate a Vue.js with Firebase to build a functional web application.
- **The things that helped me most were :** Coding exercise, debugging error message and testing scenarios
- **I found the following topics particularly challenging :** Firebase Cloud Function, Firestore rules and booking conflict detection
- **I found the following topics particularly interesting :** Achievement system, environment impact tracking
- **I feel I learnt these topics, concepts, and/or tools really well :** Firebase Authentication, cloud deployment and Vue.js components
- **I still need to work on the following areas :** Writing automated tests, improving accessibility and strengthening backend security
- **My progress in this unit was : F**rom developing basic Vue.js components to integrating multiple services and implementing more complex application features.
- **This unit will help me in the future :** Add more and more experience in web development
- **If I did this unit again I would do the following things differently :** Plan the application architecture earlier and test each feature more frequently